



Implementation and Validation of the Ethereum JSON RPC API for ERC-20 Token Transaction Retrieval and Money Laundering Detection

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Abstract

This dissertation examines the use of Ethereum's JSON-RPC API for tracing, monitoring, and extracting data of ERC-20 token transfer and analysing blockchain transactions to detect illicit activity of money laundering, within decentralized finance (DeFi) ecosystems. Blockchain technology, originally introduced to support Bitcoin, has evolved into a dynamic platform, with Ethereum leading the charge in enabling programmable smart contracts and decentralized applications (dApps). These advancements, however, come with new challenges, particularly regarding the regulation of digital assets and their potential misuse for illegal activities.

This study addresses the complexity of tracking and tracing transactions on the Ethereum blockchain, where pseudonymity and the lack of centralized control make it hard to identify suspicious transactional patterns. Existing methods for detecting suspicious transactions often rely on expensive third-party analytics services that provide advanced tracing capabilities. This research aims to determine whether comparable outcomes can be achieved using Ethereum's free, standard JSON-RPC API, thus providing a more cost-effective alternative for monitoring blockchain activities.

The dissertation details how to retrieve and decode relevant ERC-20 token transfer logs by implementing the JSON-RPC API and visualize token transfer networks to manually identify suspicious patterns without depending on external tools like QuickNode. The findings highlight the potential of using standard blockchain APIs to enhance transparency, improve regulatory compliance, and mitigate risks related with money laundering and other illegal activities in the DeFi ecosystem. By providing an affordable and scalable approach to monitoring blockchain transactions, this research highlights the value of leveraging open-access tools for ensuring financial integrity within decentralized systems.

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Abbreviations

DeFi - Decentralized Finance

dApps - Decentralized Applications

AML - Anti-Money Laundering

PoW - Proof of Work

PoS - Proof of Stake

EVM - Ethereum Virtual Machine

DAO - Decentralized Autonomous Organization

NFT - Non-Fungible Token

DEX - Decentralized Exchange

KYC - Know Your Customer

AI - Artificial Intelligence

NLP - Natural Language Processing

VPN - Virtual Private Network

E-Node - Ethereum Node

RPC - Remote Procedure Call

ABI - Application Binary Interface

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Chapter 1

1. Introduction

1.1 Introduction

The implementation of blockchain technology in recent years has revolutionized the financial services industry [1], offering new opportunities for efficiency, transparency, and decentralization [2]. However, it has also introduced a variety of new challenges [3], including the rise of illicit activities such as phishing scams [4], smurfing [5], and money laundering [4-6]. In 2022 [7] alone, an estimated \$23.8 billion worth of cryptocurrency was laundered worldwide. Blockchain platforms like Ethereum facilitate dApps [8] and financial transactions [9] using smart contracts and ERC-20 tokens [8-10], providing users with immutability and transparency. Despite these advantages, the pseudonymous nature of these transactions also presents significant risks, as adversaries can exploit the anonymity and decentralization of blockchain platforms to engage in illegal activities, including money laundering.

Money laundering [11,12] is a process that involves acquiring illegally obtained funds to make them appear legitimate. This is typically achieved by transferring funds through multiple wallets and across various platforms, effectively hiding the fraudulent activity. Traditional anti-money laundering (AML) mechanisms [13,14] rely on centralized institutions and regulatory bodies to monitor and trace illicit financial flows and are ill-suited for decentralized blockchain environments like Ethereum. The pseudonymity of blockchain transactions creates challenges for monitoring and identifying suspicious activity [14], making it difficult for regulatory bodies to effectively trace illicit financial flows.

Ethereum's ERC-20 tokens [9-10] have become a crucial standard for fungible tokens, widely used across various DeFi applications. These tokens are frequently transferred across different wallets and platforms, adding layers of complexity to the task of identifying suspicious transactions. Due to Ethereum's decentralized, token transfers occur directly between users without intermediaries, creating opportunities for malicious actors to exploit the system for illegal purposes.

To address these challenges [14,15], this project focuses on implementing Ethereum's JSON-RPC API to trace ERC-20 token transactions logs on the blockchain. Ethereum's JSON-RPC API [16] is a standard interface that allows to interact with Ethereum nodes, enabling developers to query transaction data, retrieve logs, and interact with smart contracts. By utilizing this API, the project aims to trace the flow of ERC-20 token across Ethereum smart contracts, analyse transaction patterns, and identify potentially suspicious activities such as rapid token transfers, smurfing, and large transactions to or from new or unrecognized addresses.

One of the key advantages of using Ethereum's JSON-RPC API is that it offers a cost-effective alternative [17] to expensive third-party services such as QuickNode, which are often used for advanced transaction tracing. Ethereum's JSON-RPC API provides direct access to the blockchain data, allowing for real-time analysis and monitoring of ERC-20 token transfers without the need for external. This project demonstrates that it is possible to conduct effective transaction tracing and analysis using standard, free blockchain tools, making this process accessible to smaller organisations and independent researchers.

Ultimately, this project demonstrates that Ethereum's JSON-RPC API can be an effective tool for data retrieval, monitoring token transfers and identifying potential money laundering activities within the decentralised financial ecosystem. It provides a foundational approach to tracing blockchain transactions that could be further expanded through automated analysis techniques, such as machine learning [18], to enhance the scalability and effectiveness of monitoring tools in the future.

1.2 Aim

This project aims to implement Ethereum's JSON-RPC API methods and validate the retrieved ERC-20 token transaction logs using blockchain data. Furthermore, the project seeks to analyse these transaction logs to detect and evaluate patterns potentially indicative of money laundering activities.

1.3 Objective

A list of the objectives has been compiled in accordance with the project's aim:

1. Set up a suitable development environment for implementing Ethereum's JSON-RPC API methods, including installing libraries and tools.
2. Integrate Ethereum's JSON-RPC API methods to facilitate the retrieval and processing of ERC-20 token transaction logs.
3. Analyse the transaction logs to determine token transfer movements, including sender and receiver details and the amounts transferred.
4. Validate the accuracy and completeness of the retrieved transaction data by cross referencing with QuickNode's data.
5. Visualize token transfers by illustrating transactional token movement graphs, showing relationships between wallets, tokens, and transaction patterns.
6. Identify and flag suspicious transactions based on predefined patterns and criteria indicative of potential money laundering activities.

Chapter 2

2. Background

2.1 Introduction

The literature review explores the evolution of blockchain technology with a focus on Ethereum, DeFi, and the associated challenges of money laundering. It provides critical background information, and motivations to enhance understanding of the complexities and implications of blockchain, particularly in the context of transaction monitoring and regulatory concerns.

2.2 Blockchain Technology

Blockchain technology, initially introduced as the foundational technology behind Bitcoin in 2008 [19], has rapidly evolved into a powerful tool with the potential to revolutionize a wide array of industries. From finance and supply chain management to healthcare [19,20], blockchain offers secure, transparent methods for recording transactions and managing digital assets. The Blockchain technology possess the ability to maintain a secure and immutable ledger of transactions without the need for a centralized authority. This is achieved by combining cryptographic techniques, decentralized consensus mechanisms, and distributed network architecture [19].

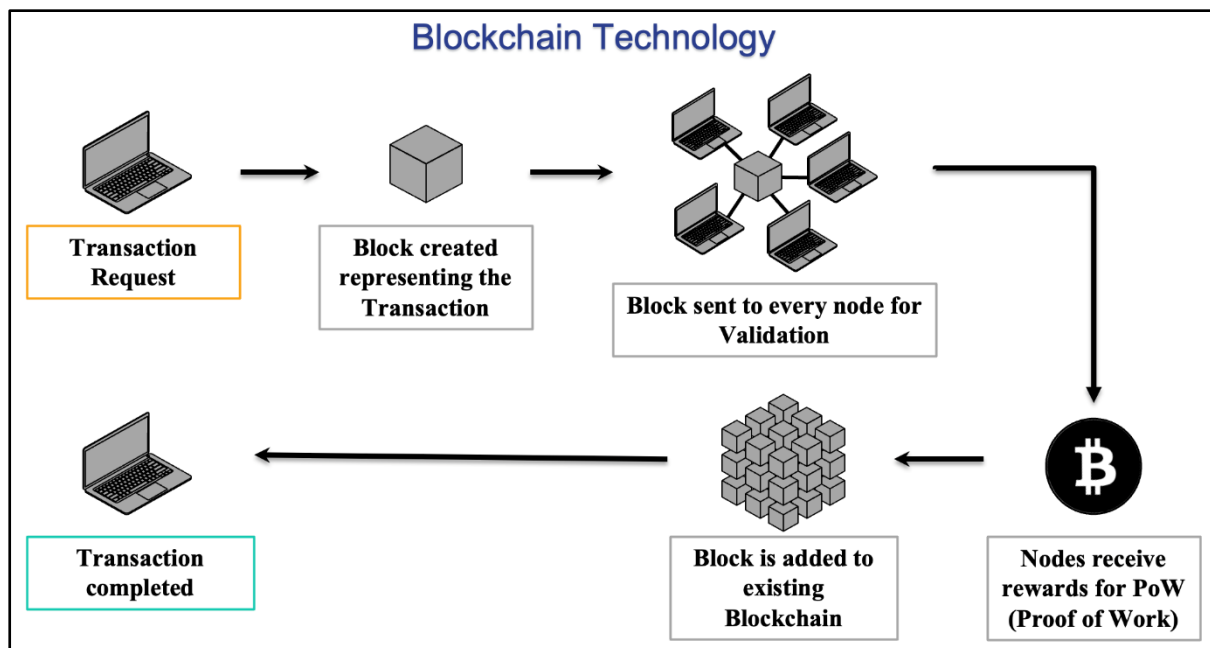


Figure 2.1: Working of Blockchain Technology

As illustrated in **Figure 2.1**, A blockchain is composed of a series of blocks, each containing a list of transactions. These blocks are cryptographically linked to the preceding block, forming a chronological chain. Every time a transaction request is made, a block representing that request is created, which is then sent to all the nodes within the blockchain network for validation. Each node in the blockchain network participates in validating these blocks of transactions through consensus mechanisms such as Proof of Work (PoW) [21] or Proof of Stake (PoS) [22]. During PoW, users, known as miners, solve complex mathematical puzzles to validate transactions and add new blocks to the chain. PoS, on the other hand, selects validators based on the amount of cryptocurrency they hold and are willing to "stake" as

collateral. Once consensus is reached, the transaction is permanently added to the blockchain's distributed ledger, ensuring it is visible and verifiable by every participant on the network.

One of the key features of blockchain technology is decentralization [22]. The traditional financial system relies on central authorities, such as banks and governments, to maintain and secure transaction records [22,23]. A blockchain, however, distributes this responsibility across all participants in the network. This decentralization creates a trustless system where transactions can occur peer-to-peer without the need for intermediaries like banks or brokers. This approach offers several benefits [24,25], including increased security, resistance to censorship, and the elimination of single points of failure. However, it also introduces challenges [26], particularly in terms of scalability and regulatory oversight.

Immutability [27] is another key characteristic of blockchain technology. Once a transaction is recorded on the blockchain, it cannot be altered or deleted. This nature of immutability is made possible using cryptographic hashes [19] that uniquely identify each block and its contents. Although immutability ensures that transaction history is accurate and cannot be manipulated, it also poses challenges [26,28], particularly in the context of regulatory compliance and data privacy.

Another fundamental principle of blockchain is transparency [29]. The public nature of blockchain transactions means that the financial activities of individuals may be exposed to the world, raising concerns about privacy and data security. This pseudonymous nature of blockchain transactions enables individuals to obfuscate their identities, leading to challenges [29] in tracking and preventing illegal activities.

2.3 Ethereum and Smart Contracts

The introduction of Ethereum blockchain technology in 2015 [30] marked a substantial progression in the domain of blockchain technology, integrating programmable smart contracts in addition to decentralized applications that surpass the functionalities offered by Bitcoin. Ethereum's flexibility and broader functionality [30,31] have made it a leading platform for blockchain innovation, enabling developers to build decentralized solutions across various industries, including DeFi, gaming, and token issuances. While Bitcoin was primarily designed as a digital currency, Ethereum's rapid ecosystem growth has expanded its reach far beyond that, supporting a wide range of applications.

Ethereum introduced the concept of smart contracts [32], which are self-executing contracts with the terms of the agreements directly written into code. Smart contracts operate on the Ethereum Virtual Machine (EVM) [33], a decentralized computing environment that ensures that all nodes in the Ethereum network execute the contract code in a decentralized manner. It eliminates the need for intermediaries to enforce contracts, resulting in transactions that are faster, efficient, and trustless transactions. These smart contracts enable the development of dApps from managing decentralized autonomous organizations (DAOs) to implementing various DeFi protocols.

The versatility of Ethereum [30] has made it the foundation for multiple decentralized applications, including decentralized exchanges, lending platforms, and non-fungible token (NFT) marketplaces. Embedding custom logic into smart contracts [32] has enabled a wide range of dApps, including those that facilitate financial transactions, identity management, or automated business processes.

However, smart contracts are not immune to risks, as their complexity can introduce vulnerabilities in code, which, if left undetected, can be exploited by malicious actors. If a smart contract is not well-audited, it can become a target for exploits, resulting in the loss of funds or data. Over the years, several notable incidents [34] have exposed security flaws in smart contracts, leading to substantial losses of funds. The infamous DAO hack in 2016 [35] exposed vulnerabilities in Ethereum's early smart contracts, resulting in a theft of over \$50 million worth of ether. Such incidents have highlighted the importance of rigorous auditing and security measures in blockchain development. Therefore, the security of smart contracts remains a crucial area of focus in blockchain development.

2.4 Decentralized Finance (DeFi)

DeFi [36] is an innovative sector within the financial industry, utilizing blockchain technology to establish an open, transparent, and permissionless financial ecosystem. DeFi applications are based on smart contracts [32,33] since these self-executing contracts eliminate the need for human intervention, enabling not only simple financial services but also significantly more complicated lending, borrowing, and trading. The automation provided by smart contracts ensures that the rules of each transaction are strictly enforced, fostering a reliable and transparent financial environment.

Decentralized exchanges (DEXs) [37] are another critical component of the DeFi ecosystem. Unlike centralized exchanges, which act as intermediaries that facilitate the buying and selling of cryptocurrencies, DEXs allow users to trade cryptocurrencies directly with one another. This peer-to-peer trading model removes the need for a central authority to manage transactions, thereby reducing the risk of censorship, hacking, and manipulation. Platforms like Uniswap and SushiSwap create a decentralized and self-sustaining market system. By allowing users to provide liquidity in exchange for a share of trading fees, DEXs facilitate a more inclusive and participatory market environment.

Stablecoins [38], a key element in the DeFi, are cryptocurrencies designed to maintain a stable value, typically pegged to a fiat currency like the US dollar. Stablecoins like DAI and USDC are crucial in DeFi [38] as they offer a stable medium of exchange and a safe harbour for investors amid the volatile cryptocurrency market. Their stability makes them integral to a wide range of DeFi applications, from trading to lending and beyond.

2.5 ERC-20 Tokens

Tokens [39] are digital assets that allow for secure and efficient transfer, storage, and verification of information and value. A common application of smart contracts is the management of tokens, which can be categorized into two types: fungible and non-fungible [40]. Fungible tokens, like currency, have properties of being divisible and interchangeable, whereas NFTs are unique and cannot be divided or exchanged on a one-to-one basis.

An advancement that has driven the growth of the Ethereum ecosystem is the ERC-20 token standard [41]. This technical standard specifies a set of rules for the creation and management of fungible tokens on the Ethereum blockchain, ensuring that these tokens can seamlessly interact with various dApps and exchanges. The ERC-20 standard provides a basic set of operations [41] that includes transferring tokens, checking the balance, and approving third-party use of tokens on behalf of the owner. Standardized protocols like this have enabled broad integrations and interoperability across the Ethereum network.

ERC-20 tokens have become one of the most important elements of the DeFi system, which includes popular tokens like USDT, USDC, and SHIB [41,42]. The interoperability introduced

by ERC-20 tokens further enhances their compatibility with wallets, exchanges, and dApps, [42] which allows them to establish a real, dynamic ecosystem that interconnects ecosystems.

Nonetheless, the benefits of ERC-20 tokens also come with significant challenges [41,42]. The ease of creating and transferring these tokens has raised concerns regarding regulatory compliance and the potential for their misuse, such as money laundering. This has led to increased scrutiny from regulators and the development of advanced tracking and monitoring tools to trace ERC-20 transactions. Furthermore, scalability issues [42] including high transactional fees and network congestion on Ethereum impact the efficiency of token transfers. To address this challenge, active research is focusing on layer-2 scaling solutions such as rollups and sidechains with the aim of improving network performance and reducing the cost of transactions.

2.6 Money Laundering

Money laundering [43] is a significant threat to the integrity of political institutions, often facilitating corruption, bribery, and the destabilization of governments. It is a global problem that involves the process of disguising the origins of illicitly obtained funds by passing them through complex financial channels to make them appear legitimate. The primary objective [44] of money laundering is to return the "cleaned" money to the launderer in an obscure and indirect way, making it difficult for authorities to trace the original source of the funds. This process traditionally follows three stages: placement, layering, and integration.

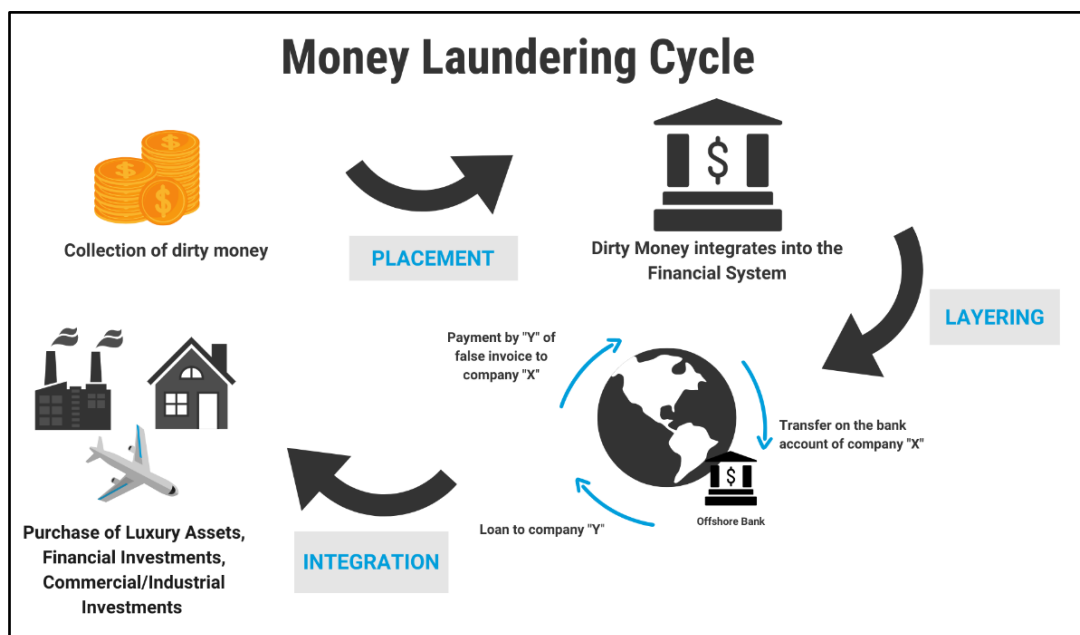


Figure 2.2: Money Laundering Cycle [45]

As illustrated in **Figure 2.2**, the first stage placement involves the introduction of illicit funds into the financial system through activities like cash deposits, purchasing assets, or buying securities. Following placement, the second stage is layering that involves a series of transactions designed to distance the funds from their illicit origin. This is often achieved by moving the money between various accounts or converting it into different forms, such as commodities, stocks, or cryptocurrencies. Layering can include complex transfers that cross international borders, making the money trail more challenging to follow. The final stage, integration, occurs when the laundered money is returned to the economy disguised as legitimate funds, often through investments in luxury assets or businesses.

Cryptocurrencies like Bitcoin and Ethereum present new challenges in combating money laundering [46]. Although public blockchains are transparent, enabling visibility of all transactions [45,46], the pseudonymous nature of these systems shields the true identities behind wallet addresses. This allows criminals to convert illicit funds into cryptocurrency (placement), transfer them across multiple wallets or use mixing services to obfuscate the trail (layering), and eventually convert them back into fiat currency or use them to purchase goods (integration).

Despite the pseudonymity, the transparency of blockchain has led to the development of advanced analytics tools designed to detect suspicious patterns and trace the flow of illicit funds. Governments and law enforcement agencies worldwide are increasingly employing these tools to combat crypto-based money laundering [43]. However, the global and decentralized nature of cryptocurrencies continues to make it challenging to fully trace these illicit activities.

2.7 Dirty Money

Dirty money [47] refers to funds generated through illegal activities, which are then laundered to appear as though they originate from legitimate sources. The blockchain's transparency, while beneficial for traceability, is coupled with a level of anonymity that makes it an attractive medium for laundering money.

Cryptocurrencies like Bitcoin, Ether, and various ERC-20 tokens can be transferred between users with minimal oversight, providing a layer of anonymity that is difficult to achieve with traditional fiat currencies. This pseudo-anonymity, combined with the global and decentralized nature of blockchain networks, makes it easier for criminals to move large sums of money across borders without detection.

The rise of DeFi has further complicated efforts to combat money laundering. Unlike centralized exchanges, which typically have KYC and AML protocols in place [47], many DeFi platforms operate without such checks. This absence of regulatory oversight allows individuals to exchange, lend, and borrow cryptocurrencies anonymously, making it easier to integrate dirty money into the legitimate financial system.

2.7.1 Smurfing

Smurfing [5] is a method used by money launderers to break down large sums of illicit funds into smaller amounts and spread them across multiple addresses. These smaller amounts are then cycled through various DeFi protocols [34-36], including lending platforms, yield farming, and decentralized exchanges. The goal is to obscure the money trail by making the transactions appear legitimate and dissociating them from their illegal origins.

Mixers or tumblers [47] are another tool frequently used in this process. While some mixers are operated with legitimate privacy concerns in mind, they are often exploited by criminals to launder dirty money.

2.8 Ethereum's JSON RPC API

Ethereum's JSON-RPC API [16-18] is an interface for developers and applications interacting with the Ethereum blockchain. JSON-RPC (JavaScript Object Notation – Remote Procedure Call) [16] is a stateless and lightweight protocol that allows for seamless communication between clients and Ethereum nodes. Through this API, users can access a variety of functions essential for blockchain operations, including querying blockchain data, managing transactions, and interacting with smart contracts.

One of the key features of the JSON-RPC API [18] is its ability to retrieve transaction details. By using the appropriate methods, developers can obtain comprehensive information about specific transactions, including the sender and receiver addresses, the transaction value, and its status. This capability is fundamental for monitoring transaction progress, verifying execution, and debugging issues. The transparency provided by these methods supports effective transaction management and helps ensure that operations on the blockchain are accurately tracked.

The API also supports interaction with smart contracts, developers can deploy, manage, and interact with these contracts through the JSON-RPC API. This functionality is vital for building dApps and executing smart contract functions. By facilitating contract interactions, the API enables the development of complex blockchain-based solutions and services.

Another significant aspect of the JSON-RPC API [17] is its ability to retrieve logs generated by smart contracts. These logs, or event logs, provide insights into the activities and events occurring within smart contracts. By querying these logs, developers can track important contract events, monitor changes, and trigger additional actions based on specific contract activities. This feature is essential for the development and maintenance of dApps, where real-time event tracking is often required.

However, the standard Ethereum JSON-RPC API does not natively support advanced tracing capabilities [16], which are often necessary for detecting complex money laundering activities. For example, while the API provides basic methods for retrieving transaction data, it does not offer built-in tools for analysing transaction flows or identifying suspicious patterns.

To address this limitation [17-18], various third-party services [48], such as QuickNode, offer enhanced tracing capabilities that allows users to track the movement of funds across multiple transactions, identify patterns such as mixing or layering, and generate detailed reports on transaction flows. These services are often costly, making them inaccessible for many users. This has led to a growing interest in developing alternative methods that can achieve similar results using the standard Ethereum JSON-RPC API, providing a more cost-effective solution for monitoring blockchain transactions. This project aims to demonstrate that similar results can be achieved using only the free, standard Ethereum JSON-RPC API.

2.9 Summary

This literature review analyses blockchain technology's evolution, focusing on its impact, particularly through Ethereum and associated challenges like money laundering. Blockchain technology provides a decentralized, secure, and immutable ledger for recording transactions, eliminating the need for central authorities through consensus mechanisms like PoW and PoS. This innovation ensures transaction security and transparency but introduces challenges in scalability, regulatory compliance, and privacy.

Ethereum, launched in 2015, advanced blockchain technology by introducing programmable smart contracts and dApps. Smart contracts, running on the EVM, execute code automatically, facilitating trustless transactions and supporting a broad range of applications including DeFi and NFTs. Despite its flexibility, Ethereum faces risks from complex smart contracts that can be vulnerable to exploits.

The ERC-20 token standard has significantly contributed to Ethereum's growth by providing a framework for fungible tokens, enhancing interoperability among dApps and exchanges. However, ERC-20 tokens face regulatory and scalability issues, prompting research into solutions like rollups and sidechains to address high transaction fees and network congestion.

Money laundering, a global issue involving disguising illicit funds, is complicated by cryptocurrencies due to their pseudonymous nature. While blockchain's transparency aids visibility, tracking illicit funds remains challenging. Techniques such as smurfing and mixers obscure transaction trails, complicating anti-money laundering efforts. The Ethereum JSON-RPC API supports basic blockchain interactions but lacks advanced tracing capabilities, leading to the development of costly third-party services. The review aims to explore whether similar tracing can be achieved using the free standard API.

Chapter 3

3. Related work

3.1 Evolution of Money Laundering

In the 1970s, financial institutions started reporting transactions to government authorities to detect money laundering [49]. The regulatory framework and technology used to combat money laundering have evolved as these illicit activities became more prevalent and sophisticated.

At the beginning, automated banking systems employed rudimentary rule-based detection mechanisms. These systems flagged transactions based on certain thresholds, such as substantial cash deposits or frequent international fund transfers. Savage et al. [50] introduced an innovative system that integrated network analysis and supervised learning techniques to identify suspicious groups of customers. In their approach, they focused on transaction types that pose a high risk of money laundering, such as large deposits of cash and international wire transfers. This early approach laid the groundwork for advanced transaction monitoring systems.

3.2 Machine Learning and Advanced Techniques

Detection of financial crime has become increasingly sophisticated because of the advancement of methodologies. Martínez-Sánchez et al. [51] further extended this by leveraging historical transaction data and tree regression techniques. Based on various customer-specific predictors, such as legal entity type, source of funds, economic activity, and customer seniority, their model predicted money laundering risks. These variables allowed the model to assess the risk levels associated with specific customers, facilitating more targeted AML measures.

The study conducted by Jullum et al. [52] also developed a predictive model for forecasting whether new transactions will require reporting. Moreover, their model included information on sender and receiver, past behaviour, and transaction history, enabling them to identify nuanced patterns likely to evade simpler rule-based systems.

In Domashova and Mikhailina's study [53], key factors used in machine learning models for money laundering detection were identified, such as an organization's type, age, authorized capital, founders, taxes paid, and profits generated. As a result of these variables, money laundering detection systems have become more accurate in identifying organizations with unusual or suspicious financial profiles. Despite these challenges, Rosha-Salazar et al. [43] note that there is a shortage of historical money laundering cases, which limits the effectiveness of traditional supervised learning techniques.

3.3 Recent Advances in AML Regulations and Policies

In response to money laundering's evolving nature, recent regulatory frameworks have placed a significant emphasis on compliance. Farber [54] discussed new policies designed to stem terrorism financing through the freezing of financial assets connected to terrorism. As a result of these policies, criminal and civil provisions are now in effect targeting banks and financial institutions that manage accounts associated with terrorism.

Tommandru and Chakka [55] highlighted the importance of compliance in AML systems, particularly in integrating Know Your Customer (KYC) data into unified repositories. Through this integration, a more holistic view of the customer and their activities can be obtained, which is critical when dealing with decentralized financial systems such as Ethereum.

The research conducted by Gaviyau et al. [56] showed the financial penalties institutions face when they commit repeated legal violations and fail to take corrective measures. These studies highlight the financial and reputational risks associated with inadequate AML systems, thus reinforcing the importance of retrieving and monitoring transactions in an accurate and reliable manner

3.4 Money Laundering Detection in the Ethereum Ecosystem

Based on Ethereum's transparency, Chen et al. [57] developed a framework using graph-based techniques to identify suspicious transactions. By using JSON-RPC, they queried blockchain data, analysed transaction flows, and identified abnormal patterns indicative of money laundering. The researchers developed a model that could flag unusual clusters of transactions associated with money laundering by tracking Ethereum's smart contract interactions.

Furthermore, Fan et al. [58] developed an automated tool to audit smart contracts utilizing Ethereum's JSON-RPC interface. As part of their work, they evaluated the behaviour of smart contracts deployed on the Ethereum network to detect fraudulent schemes, such as Ponzi schemes, which are frequently used to launder money. With the help of the JSON-RPC method and continuous monitoring of transactions, the researchers were able to identify contract patterns that differed from the usual DeFi protocols.

3.5 Recent research on money laundering

In recent years, research on money laundering has focused on the use of advanced technologies such as machine learning, blockchain analysis, and artificial intelligence (AI) [12] to improve detection and prevention mechanisms. One of the significant developments is the integration of machine learning models for the detection of suspicious transactions. The study conducted by Zafarullah et al. [59] analyses transaction patterns in order to identify irregularities that may indicate money laundering activities. These approaches offer greater precision than traditional rule-based systems, particularly when it comes to detecting sophisticated laundering methods such as layering and smurfing.

Another area of interest has been the use of blockchain technology and cryptocurrency in money laundering schemes, particularly with the rise of DeFi. By using graph-based methods, Gao et al. [60] examined techniques for tracking illicit cryptocurrency transactions using blockchain analytics. Their research addresses the challenges posed by the pseudonymity and decentralization of blockchains, offering new methods for identifying laundering patterns in digital currencies.

Furthermore, Chakraborty et al. [61] emphasized the importance of natural language processing (NLP) in the analysis of financial documents, including KYC records and suspicious activity reports (SARs). In their research, they demonstrate how NLP can automate the extraction of critical information from vast quantities of unstructured data, thereby improving the efficiency of compliance teams in identifying money laundering risks.

Nguyen et al. [62] examined the impacts of global AML regulations on financial institutions, with a particular focus on enhancing cross-border cooperation. Their study highlights the importance of public-private partnerships and information-sharing protocols in improving the global fight against money laundering, especially in addressing complex, transnational laundering networks.

In conclusion, recent advances in money laundering detection have seen a shift towards utilizing cutting-edge technologies such as machine learning, blockchain analytics, and AI. These innovations are designed to address the complexity of modern laundering techniques, particularly in decentralized environments such as cryptocurrency and digital financial services. Combining deep learning algorithms, graph-based blockchain analysis, and NLP enhances the ability to detect subtle patterns and hidden networks.

Chapter 4

4. Methodology

4.1 Introduction

This chapter details the comprehensive process of setting up a development environment for Ethereum blockchain interaction and visualization. It begins with the installation of essential tools and libraries. Following this setup, the process includes connecting to an Ethereum node and integrating Ethereum's JSON RPC API methods for extracting and decoding transaction data. The chapter then covers the setup for data validation to ensure accuracy. Finally, it describes the visualization of token transfer networks, providing insights through graphical representation of the data.

4.2 Development Environment Setup

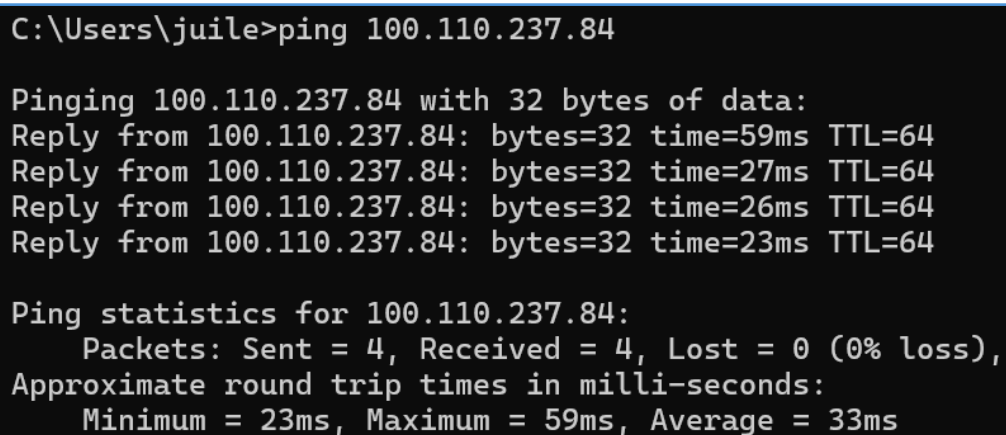
The first step in this chapter is establishing a development environment to interact with the Ethereum blockchain. This requires installing the tools and dependencies, including Python, Web3.py, Plotly, and Dash Cytoscape. These libraries are critical for data retrieval, processing, and visualization.

- **Python Installation:** Python Version 3.8 is installed to ensure compatibility with Web3.py and other libraries such as pandas, dash. Python is being selected as the primary programming language due to its extensive libraries.
- **Web3.py Installation:** Web3.py is installed using `pip install web3` command to interact with Ethereum blockchain nodes. This library allows retrieval of blockchain data and facilitates Ethereum transactions.

Connection to the Ethereum Node (E-Node)

The next step involves connecting to an E-Node provided by supervisor, Dr. Pascal Berrang. This E-Node is accessible via a Virtual Private Network (VPN) connection through Tailscale.

- **Tailscale Setup:** Tailscale is a mesh VPN tool that is allowing secure connections to private nodes. After creating an account on Tailscale, a secure tunnel is being established between local development environment and the E-Node.
- **Node Ping Test:** The E-Node is pinged at IP address **100.110.237.84** to verify the connection as visible in the **Figure 4.1**. This step confirmed the accessibility of the node before proceeding with blockchain interaction.

A terminal window with a black background and white text. The command 'C:\Users\juile>ping 100.110.237.84' is entered. The output shows four successful replies from 100.110.237.84 with decreasing times (59ms, 27ms, 26ms, 23ms) and a TTL of 64. Ping statistics show 4 packets sent, 4 received, 0% loss, and average round trip times of 33ms.

```
C:\Users\juile>ping 100.110.237.84

Pinging 100.110.237.84 with 32 bytes of data:
Reply from 100.110.237.84: bytes=32 time=59ms TTL=64
Reply from 100.110.237.84: bytes=32 time=27ms TTL=64
Reply from 100.110.237.84: bytes=32 time=26ms TTL=64
Reply from 100.110.237.84: bytes=32 time=23ms TTL=64

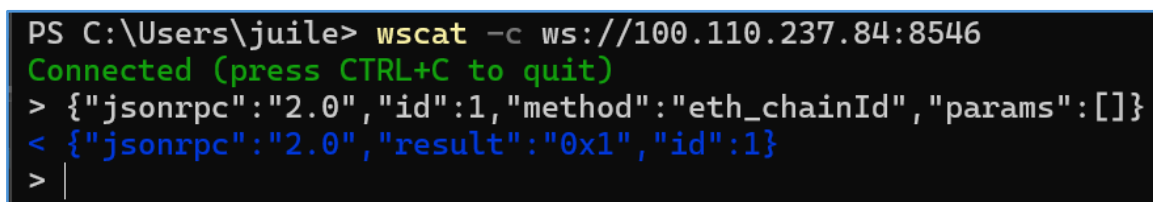
Ping statistics for 100.110.237.84:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 59ms, Average = 33ms
```

Figure 4.1: Connection verification by pinging node

- **Node.js, NPM and wscat Setup:** Node.js and npm are installed to use the WebSocket client wscat, which is essential for interacting with Ethereum in real time.
 - Installing wscat using the command `npm install -g wscat`.
 - Establishing a WebSocket connection to the E-Node using the command `wscat -c ws://100.110.237.84:8546`.
 - After connecting successfully, the first command sent is:


```
{"jsonrpc": "2.0", "id": 1, "method": "eth_chainId", "params": []}
```

As shown in **Figure 4.2**, Ethereum chain ID is verified using this command whereas result of "0x1" is indicated that the connection is successfully established to the E-Node.



```
PS C:\Users\juile> wscat -c ws://100.110.237.84:8546
Connected (press CTRL+C to quit)
> {"jsonrpc":"2.0","id":1,"method":"eth_chainId","params":[]}
< {"jsonrpc":"2.0","result":"0x1","id":1}
> |
```

Figure 4.2: *WebSocket Connection to E-Node*

4.3 Integration of Ethereum's JSON RPC API Methods

Once connected to the E-Node, Ethereum's standard JSON RPC API methods are utilized to retrieve transaction data. This involves using specific methods to gather detailed transaction information, decode ERC-20 token transfers, and extract relevant data fields such as sender and receiver addresses, token amounts, token symbols and contract addresses.

4.3.1 JSON RPC Methods Used

The Ethereum Remote Procedure Call (RPC) interface is employed to extract data using three API calls: `eth_getLogs`, `eth_getTransactionReceipt`, and `eth_getTransactionByHash` [16]. These methods are crucial for obtaining detailed transaction details.

- **eth_getTransactionByHash:** This method retrieves detailed information about a specific transaction by its unique identifier (hash). The transaction hash serves as a reference to obtain transaction data such as sender address, receiver address, and transaction value.
- **eth_getTransactionReceipt:** This method confirms the completion of a transaction and provides the block number in which the transaction is included. Additionally, it indicates whether the transaction succeeds or fails.
- **eth_getLogs:** This method is used to filter as well as retrieve logs related to ERC-20 token transfers. Since ERC-20 tokens emit the *Transfer* event each time tokens are transferred, the `eth_getLogs` method is essential to capturing these events.

The following **Figure 4.3** demonstrates the process of retrieving logs for a specific block.

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "eth_getLogs",
  "params": [
    {
      "fromBlock": "0x429d3b",
      "toBlock": "0x429d3b",
      "address": "0xb59f67a8bffa5d8cd03f6ac17265c550ed8f33907",
      "topics": [
        "0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef",
        "0x00000000000000000000000000000000b46c2526e227482e2ebb8f4c69e4674d262e75",
        "0x0000000000000000000000000000000054a2d42a40f51259dedd1978f6c118a0f0eff078"
      ]
    }
  ]
}
```

Figure 4.3: JSON RPC Request

This query captures all *Transfer* events that occurred within the specified block for a particular ERC-20 token. The log data returned provides detailed information about token transfers, including sender and receiver addresses, token amounts, and contract addresses.

4.3.2 ABI Fragments for ERC-20 Tokens

To decode and process the raw log data retrieved via the Ethereum JSON RPC API, Application Binary Interface (ABI) fragments are required. These ABI fragments allow interaction with specific ERC-20 token contract methods such as name, symbol, and decimals. The following **Figure 4.4** demonstrates how ABI fragment interacts with these contract methods.

```

# ABI fragments for the ERC20 `name`, `symbol`, and `decimals` methods
ERC20_ABI = [
    {
        "constant": True,
        "inputs": [],
        "name": "name",
        "outputs": [{"name": "", "type": "string"}],
        "type": "function",
    },
    {
        "constant": True,
        "inputs": [],
        "name": "symbol",
        "outputs": [{"name": "", "type": "string"}],
        "type": "function",
    },
    {
        "constant": True,
        "inputs": [],
        "name": "decimals",
        "outputs": [{"name": "", "type": "uint8"}],
        "type": "function",
    }
]

```

Figure 4.4: ABI fragments

4.3.3 Decoding Logs

The raw log data returned by the `eth_getLogs` method is decoded using Python scripts to extract human-readable details. The `decode_logs` function is implemented to parse the logs, translating hexadecimal data into understandable information such as the addresses involved in transactions as well as the amounts transferred.

4.4 Validation of Logs

To ensure the accuracy and reliability of the data retrieved from the E-Node, a validation step is introduced using QuickNode.

QuickNode Setup

- **QuickNode Account Creation:** A free-tier account is created on QuickNode to gain access to Ethereum's mainnet via both HTTP and WebSocket providers.
- **Connection to QuickNode:** QuickNode provides unique URLs for connecting to their nodes. The following URLs are used:
 - HTTP Provider:
<https://long-purple-sponge.quiknode.pro/c26d5842f372ee6e32fca748626397f1fa767b5c/>
 - WebSocket Provider:
<wss://long-purple-sponge.quiknode.pro/c26d5842f372ee6e32fca748626397f1fa767b5c/>

As the given E-Node uses WebSocket for connection, thus using WebSocket provider, a Python script is written to validate the transaction data retrieved from the E-Node by comparing it to the data from QuickNode. The validation is focusing on key transaction attributes such as sender and receiver addresses, token amounts, token symbol and contract addresses. Matching data from both nodes ensures the accuracy of the retrieved blockchain data.

4.5 Visualization of Token Transfer Networks

After data retrieval and validation, the next step is to visualize the ERC-20 token transactions graphically. Plotly and Dash Cytoscape are used for creating interactive graphs and network visualizations of token flows.

- **Plotly and Dash Cytoscape Installation:** Plotly and Dash Cytoscape are installed using command `pip install plotly dash-cytoscape`. Plotly is a graphing library used for creating dynamic graphs, while Dash Cytoscape is an interactive tool for visualizing networks, such as token transfer networks. These libraries are enabling the creation of graphical representations of ERC-20 token transfers, enhancing the interpretation of the results.
- **Graph Creation Using Dash Cytoscape:** Dash Cytoscape, an extension of the Dash framework, is utilized to create network visualizations. Cytoscape visualizes complex networks like token transfer networks, where nodes represent Ethereum addresses, and edges represent token transfers.
- **Transaction Data Output to Excel:** A python script is written to export the retrieved transaction logs into an Excel sheet. The Excel sheet contains detailed data such as transaction hashes, sender and receiver addresses, and token amounts along with token symbols. This data is then used as input for the Plotly graphing tool.

4.6 Discussion

This chapter outlines the process of setting up a development environment for interacting with the Ethereum blockchain. Initially, Python 3.8 is installed, followed by Web3.py, Plotly, and Dash Cytoscape to facilitate blockchain data retrieval, processing, and visualization. The connection to an E-Node provided by Dr. Pascal Berrang is established via Tailscale, a VPN tool, ensuring secure access. Node.js and the WebSocket client wscat are set up to enable real-time Ethereum interaction, with successful connection verification through Ethereum chain ID retrieval.

JSON RPC API methods such as `eth_getTransactionByHash`, `eth_getTransactionReceipt`, and `eth_getLogs` are used to extract transaction data, including ERC-20 token transfers. ABI fragments are employed to decode this data into readable format. To validate the accuracy of the retrieved data, a QuickNode account is created, and data from the E-Node is compared with QuickNode's data. For visualization, Plotly and Dash Cytoscape are used to create interactive graphs representing ERC-20 token transfer networks.

Chapter 5

5. Implementation and Results

5.1 Introduction

This chapter presents the results obtained from executing four scripts designed to interact with the Ethereum blockchain. These scripts were developed to extract, validate, and visualize transaction data of ERC-20 token transfers. The outputs include transaction data logs, validation comparisons, data exportation to Excel, graphical visualizations of token transfer networks, and the identification of potentially suspicious transactions of money laundering.

5.2 Transaction Data Extraction

The first significant output involves the extraction of detailed transaction logs from the Ethereum blockchain. This was achieved using the Ethereum JSON-RPC API methods, specifically `eth_getTransactionByHash` and `eth_getLogs`.

The primary script developed for this task successfully connects to the E-Node and retrieves the necessary transaction data. The output of this script, as captured in **Figure 5.1**, includes the following key details:

- **Transaction Hash:** A unique identifier for each transaction, essential for tracking and referencing specific transactions on the blockchain.
- **Sender and Receiver Addresses:** The Ethereum addresses of sender and receiver involved in the transaction. These are critical for understanding who initiated the transaction and who was the recipient.
- **Token Amounts:** The number of ERC-20 tokens transferred in the transaction. This data is vital for quantifying the transaction's scale and understanding the value transferred.
- **Token Symbols:** Token symbols identify the specific type of token being transferred, providing insight into the nature of the transaction.
- **Token Contract Address:** The address of the ERC-20 token contract, which specifies the token involved in the transaction.

These details were successfully decoded from the raw blockchain data using ABI fragments. The ABI allows the script to interpret the hexadecimal data returned by the blockchain, translating it into human-readable information. This process confirms the script's effectiveness in accessing and interpreting blockchain data, ensuring that all necessary details are accurately captured.

```
Command Prompt
C:\Users\juile\OneDrive\Desktop\Dissertation Updates\Programs\Final program>python Transaction_Tracing.py

1. Tracing transaction: 0x4606654aaf02a0c69b57e10d7ddc0cbd98d23e5d681cfe60a5a415cc4ba04986
Transfer from 0x36a8dCdbF5F1Fdbf559b79a3fBb3791EEDeDC695 to 0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43 of
5294.57 BADGER (Badger) tokens from Contract 0x3472A5A71965499acd81997a54BBA8D852C6E53d

2. Tracing transaction: 0x9820fce81e5ad1a7d963be407bceadeb5d000a07a8b08d468030748c413e6774
Transfer from 0x1980475c3a279935Ca611c9A41f17974ebC2e050 to 0x28104d4F703EE5B5011cefe106f54eFd56F33f95 of
71947.988805 USDC (USD Coin) tokens from Contract 0xA0b86991c6218b36c1d19D4a2e9Eb0cE3606eB48
Transfer from 0xC9f93163c99695c6526b799EbcA2207Fdf7D61aD to 0x28104d4F703EE5B5011cefe106f54eFd56F33f95 of
71999.636973 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7
Transfer from 0x28104d4F703EE5B5011cefe106f54eFd56F33f95 to 0xC9f93163c99695c6526b799EbcA2207Fdf7D61aD of
71947.988802 USDC (USD Coin) tokens from Contract 0xA0b86991c6218b36c1d19D4a2e9Eb0cE3606eB48
Transfer from 0xC9f93163c99695c6526b799EbcA2207Fdf7D61aD to 0x95C4F5b83aA70810D4f142d58e5F7242Bd891CB0 of
0.719996 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7
Transfer from 0x28104d4F703EE5B5011cefe106f54eFd56F33f95 to 0xCf5540fFFCdC3d510B18bFcA6d2b9987b0772559 of
71999.636973 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7
Transfer from 0xCf5540fFFCdC3d510B18bFcA6d2b9987b0772559 to 0x1980475c3a279935Ca611c9A41f17974ebC2e050 of
71999.636967 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7

3. Tracing transaction: 0xf0ca92d76d405286b4aa231e3685b1057745b2b59d05b86335511d9aa4a9e34f
Transfer from 0x21a31Ee1afC51d94C2eFcCAa2092aD1028285549 to 0x36a8dCdbF5F1Fdbf559b79a3fBb3791EEDeDC695 of
5294.5 BADGER (Badger) tokens from Contract 0x3472A5A71965499acd81997a54BBA8D852C6E53d

4. Tracing transaction: 0xae6a8a9dda931fe8eeac8f194723d57cf970c546c04a0c16f216bfc8f1c71d20
Transfer from 0x36a8dCdbF5F1Fdbf559b79a3fBb3791EEDeDC695 to 0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43 of
5303.61 BADGER (Badger) tokens from Contract 0x3472A5A71965499acd81997a54BBA8D852C6E53d

C:\Users\juile\OneDrive\Desktop\Dissertation Updates\Programs\Final program>
```

Figure 5.1: Command Prompt Output

Figure 5.1 further provides a command prompt output showing the result of a Python script used to trace specific cryptocurrency transactions using Transaction hash. The script outputs detailed information about the flow of USDC (USD Coin) and USDT (Tether USD) tokens between addresses. Each transfer is listed with the source and destination addresses, along with the amount transferred, allowing for precise tracking of token movements. The output reveals that the script successfully traced multiple transfer logs involving significant amounts of USDC and USDT.

5.3 Validation of Transaction Data

To ensure the accuracy and reliability of the transaction data retrieved from the Ethereum blockchain, a validation step was implemented. This method involved comparing the data obtained from the E-Node with the same data retrieved from QuickNode. The purpose of this comparison is to verify that the data is consistent across different nodes, thereby confirming its accuracy.

The output, as shown in the Figure 5.2 & 5.3, includes the following validations:

- **Matching Transaction Hashes:** The transaction hash is the primary identifier for transactions on the blockchain. The matching hashes confirm that both nodes recognize and identify the same transaction.

- **Consistent Token Amounts:** The amounts of tokens transferred in the transaction are compared between the two data sources. Consistent token amounts across both nodes indicate that the data is accurate.
- **Matching Sender and Receiver Addresses:** The Ethereum addresses of the sender and receiver are also compared. Matching addresses confirm that the transaction flow is correctly captured by both nodes.

```

Command Prompt
1. Validating transaction 0x4606654aaf02a0c69b57e10d7ddc0cbd98d23e5d681cfe60a5a415cc4ba04986

Fetching logs from QuickNode
Transfer from 0x36a8dCDBF5F1Fdbf559b79a3fBb3791EEDeDC695 to 0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43
of 5294.57 BADGER (Badger) tokens from Contract 0x3472A5A71965499acd81997a54BBA8D852C6E53d

Fetching logs from E-Node
Transfer from 0x36a8dCDBF5F1Fdbf559b79a3fBb3791EEDeDC695 to 0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43
of 5294.57 BADGER (Badger) tokens from Contract 0x3472A5A71965499acd81997a54BBA8D852C6E53d

All log entries are validated for transaction hash 0x4606654aaf02a0c69b57e10d7ddc0cbd98d23e5d681cfe60a5
a415cc4ba04986

2. Validating transaction 0x9820fce81e5ad1a7d963be407bceadeb5d000a07a8b08d468030748c413e6774

Fetching logs from QuickNode
Transfer from 0x1980475c3a279935Ca611c9A41f17974ebC2e050 to 0x28104d4F703EE5B5011cefe106f54eFd56F33f95
of 71947.988805 USDC (USD Coin) tokens from Contract 0xA0b86991c6218b36c1d19D4a2e9Eb0cE3606eB48
Transfer from 0xC9f93163c99695c6526b799EbcA2207Fdf7D61aD to 0x28104d4F703EE5B5011cefe106f54eFd56F33f95
of 71999.636973 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7
Transfer from 0x28104d4F703EE5B5011cefe106f54eFd56F33f95 to 0xC9f93163c99695c6526b799EbcA2207Fdf7D61aD
of 71947.988802 USDC (USD Coin) tokens from Contract 0xA0b86991c6218b36c1d19D4a2e9Eb0cE3606eB48
Transfer from 0xC9f93163c99695c6526b799EbcA2207Fdf7D61aD to 0x95C4F5b83aA70810D4f142d58e5F7242Bd891CB0
of 0.719996 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7
Transfer from 0x28104d4F703EE5B5011cefe106f54eFd56F33f95 to 0xCf5540FFFCdC3d510B18bFcA6d2b9987b0772559
of 71999.636973 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7
Transfer from 0xCf5540FFFCdC3d510B18bFcA6d2b9987b0772559 to 0x1980475c3a279935Ca611c9A41f17974ebC2e050
of 71999.636967 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7

Fetching logs from E-Node
Transfer from 0x1980475c3a279935Ca611c9A41f17974ebC2e050 to 0x28104d4F703EE5B5011cefe106f54eFd56F33f95
of 71947.988805 USDC (USD Coin) tokens from Contract 0xA0b86991c6218b36c1d19D4a2e9Eb0cE3606eB48
Transfer from 0xC9f93163c99695c6526b799EbcA2207Fdf7D61aD to 0x28104d4F703EE5B5011cefe106f54eFd56F33f95
of 71999.636973 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7
Transfer from 0x28104d4F703EE5B5011cefe106f54eFd56F33f95 to 0xC9f93163c99695c6526b799EbcA2207Fdf7D61aD
of 71947.988802 USDC (USD Coin) tokens from Contract 0xA0b86991c6218b36c1d19D4a2e9Eb0cE3606eB48
Transfer from 0xC9f93163c99695c6526b799EbcA2207Fdf7D61aD to 0x95C4F5b83aA70810D4f142d58e5F7242Bd891CB0
of 0.719996 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7
Transfer from 0x28104d4F703EE5B5011cefe106f54eFd56F33f95 to 0xCf5540FFFCdC3d510B18bFcA6d2b9987b0772559
of 71999.636973 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7
Transfer from 0xCf5540FFFCdC3d510B18bFcA6d2b9987b0772559 to 0x1980475c3a279935Ca611c9A41f17974ebC2e050
of 71999.636967 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7

All log entries are validated for transaction hash 0x9820fce81e5ad1a7d963be407bceadeb5d000a07a8b08d4680
30748c413e6774

```

Figure 5.2: Command Prompt Output for Transaction Validation 1

```
Command Prompt
Transfer from 0xCf5540FFCdC3d510B18bFcA6d2b9987b0772559 to 0x1980475c3a279935Ca611c9A41f17974ebC2e050 of 71999.63696
7 USDT (Tether USD) tokens from Contract 0xdAC17F958D2ee523a2206206994597C13D831ec7

All log entries are validated for transaction hash 0x9820fce81e5ad1a7d963be407bceadeb5d000a07a8b08d468030748c413e6774

3. Validating transaction 0xf0ca92d76d405286b4aa231e3685b1057745b2b59d05b86335511d9aa4a9e34f

Fetching logs from QuickNode
Transfer from 0x21a31Ee1afC51d94C2eFcCAa2092aD1028285549 to 0x36a8dCdbF5F1Fdbf559b79a3fBb3791EEDeDC695 of 5294.5 BADG
ER (Badger) tokens from Contract 0x3472A5A71965499acd81997a54BBA8D852C6E53d

Fetching logs from E-Node
Transfer from 0x21a31Ee1afC51d94C2eFcCAa2092aD1028285549 to 0x36a8dCdbF5F1Fdbf559b79a3fBb3791EEDeDC695 of 5294.5 BADG
ER (Badger) tokens from Contract 0x3472A5A71965499acd81997a54BBA8D852C6E53d

All log entries are validated for transaction hash 0xf0ca92d76d405286b4aa231e3685b1057745b2b59d05b86335511d9aa4a9e34f

4. Validating transaction 0xae6a8a9dda931fe8eeac8f194723d57cf970c546c04a0c16f216bfc8f1c71d20

Fetching logs from QuickNode
Transfer from 0x36a8dCdbF5F1Fdbf559b79a3fBb3791EEDeDC695 to 0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43 of 5303.61 BAD
GER (Badger) tokens from Contract 0x3472A5A71965499acd81997a54BBA8D852C6E53d

Fetching logs from E-Node
Transfer from 0x36a8dCdbF5F1Fdbf559b79a3fBb3791EEDeDC695 to 0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43 of 5303.61 BAD
GER (Badger) tokens from Contract 0x3472A5A71965499acd81997a54BBA8D852C6E53d

All log entries are validated for transaction hash 0xae6a8a9dda931fe8eeac8f194723d57cf970c546c04a0c16f216bfc8f1c71d20

C:\Users\juile\OneDrive\Desktop\Dissertation Updates\Programs\Final program>
```

Figure 5.3: Command Prompt Output for Transaction Validation 2

Figure 5.2 & 5.3 illustrates the process of validating transactions using a Python script. The terminal output displays the tracking of transactions involving tokens such as Tether USD (USDT) and USD Coin (USDC), along with their corresponding contract addresses and amounts transferred. The program follows the transaction hash and confirms the transfer of funds between different wallet addresses and contracts, as demonstrated by the various transaction traces and token transfers. The final output verifies that all log entries have been successfully validated, ensuring the legitimacy and accuracy of the recorded transactions.

5.4 Export of Transaction Data to Excel

Following the successful extraction and validation of transaction data, the next step involved exporting this data to an Excel sheet. This process was designed to organize the data into a structured format, making it easier to analyse and visualize.

The script developed for this task takes the validated transaction data and systematically organizes it into columns within an Excel file. The exported data includes columns such as Transaction Hash, Sender and Receiver Addresses, Token Amounts, Token Symbols, Contract Addresses, Block Number.

The export process also includes a verification step to ensure that all relevant transaction details are accurately captured and organized in the Excel sheet. This structured data is essential for the subsequent creation of graphs and visual representations of the token transfer networks.

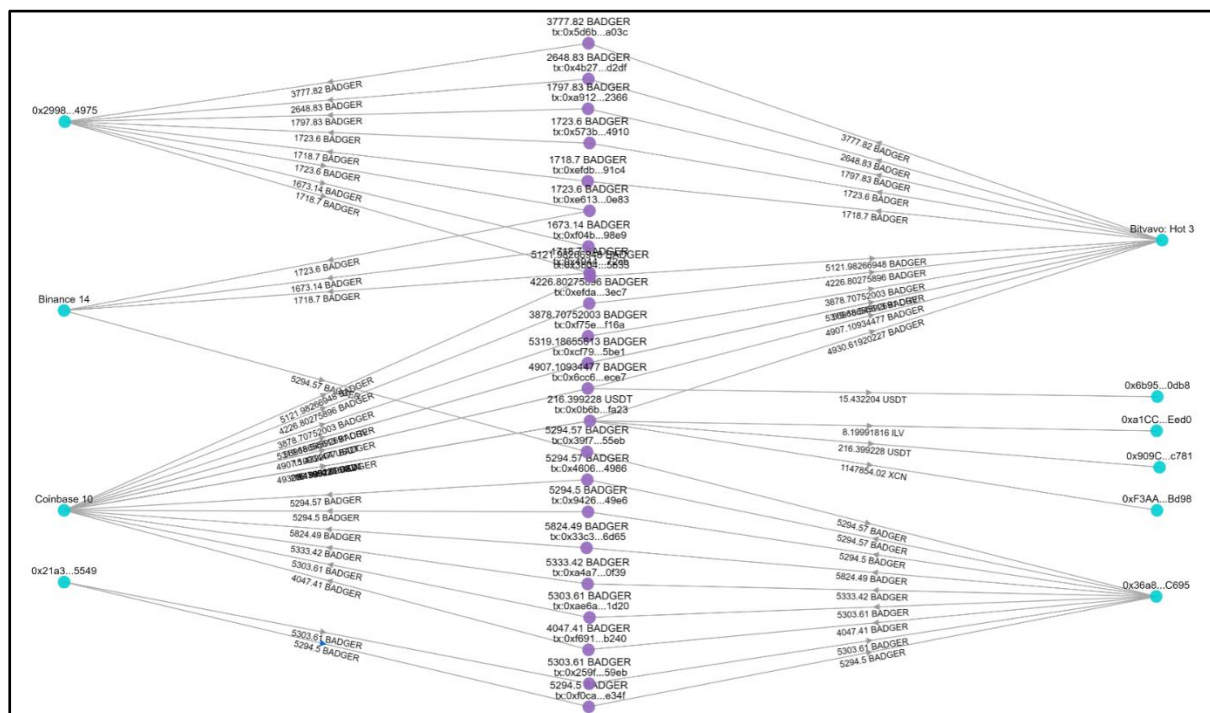
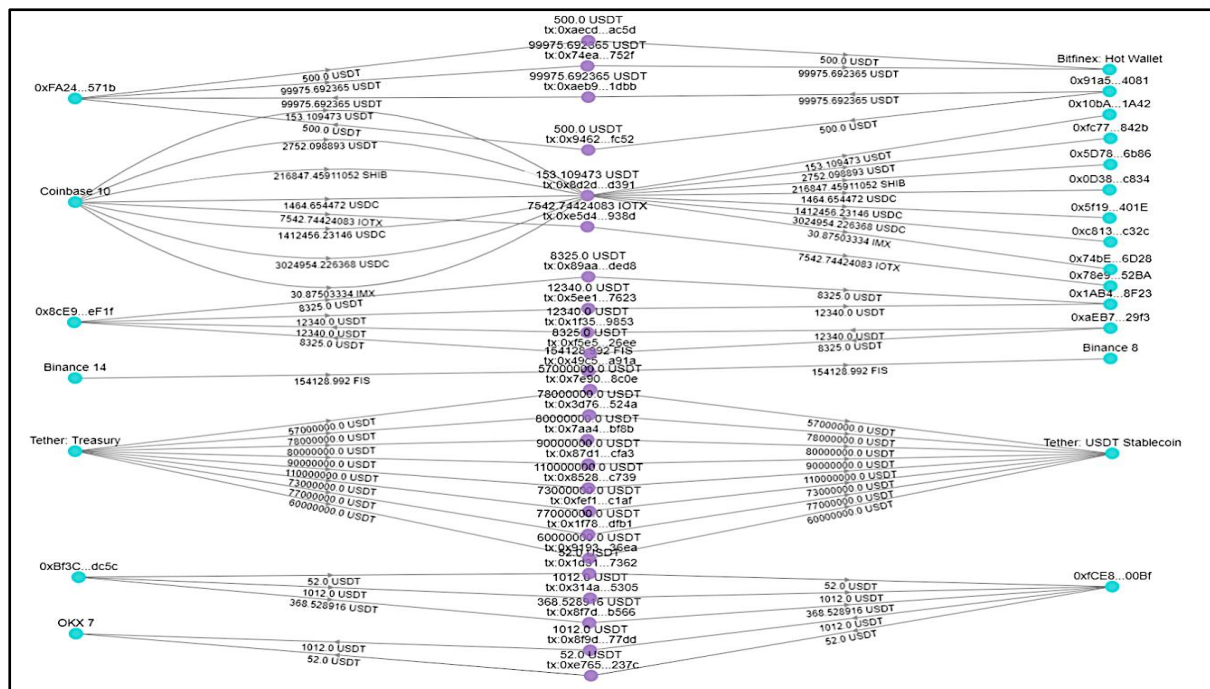


Figure 5.5 and 5.6 illustrates a network graph representing the flow of transactions between different addresses. Each node in the network represents a unique address, and the edges denote the transactions between them. The network is centralized around key addresses such as Coinbase and Binance, with numerous transactions involving the BADGER token. The high density of connections surrounding these central nodes suggests that these addresses play a significant role in the ecosystem of the BADGER token, potentially acting as large liquidity providers or exchange points. This network view helps to identify clusters of activity, pinpointing addresses that warrant further investigation due to their high transaction volumes or their central role in the network.

5.6 Identification of Suspicious Token Transfers

The last result outcome from this project involves the identification of potentially suspicious token transfers within the Ethereum network. This task was crucial for detecting potentially fraudulent activities that might be occurring within the blockchain. The process of identification relied on analysing the patterns manually and characteristics of token transfers, with a focus on the following criteria:

- **Unusually Large Transfers:** Transactions involving a significantly higher volume of tokens than usual. These transfers could indicate market manipulation, money laundering, or other illicit activities.
- **Frequent Transfers Between Specific Addresses:** Repeated transactions between the same addresses could indicate an attempt to obscure the origin or destination of tokens, a common tactic used in money laundering or other fraudulent schemes.

The suspicious transactions identified through manual examination are represented in **Figure 5.7 and 5.8**, providing a visual reference for further investigation. This output is particularly valuable for entities involved in blockchain security, compliance, or regulatory monitoring, as it helps pinpoint transactions that may require closer scrutiny.

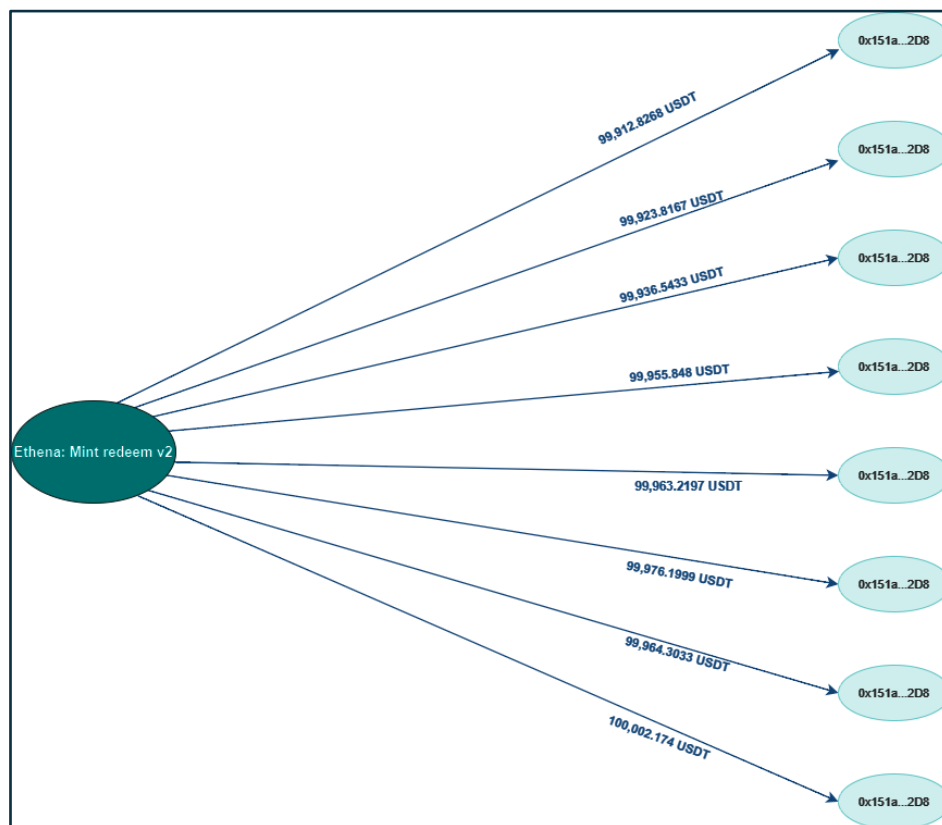


Figure 5.7: Suspicious Token Transfer Pattern 1

In the **Figure 5.7**, the pattern appears to depict a transaction flow involving the "Ethena: Mint redeem v2" entity. This entity is performing multiple transactions, where it sends specific amounts of USDT (Tether, a stablecoin) to a single recipient address identified as "0x151a...2D8." The diagram shows eight separate transactions, each with varying amounts of USDT, all flowing into this single recipient address.

This pattern could be suspicious for money laundering as it involves a large number of transactions that may be structured in such a way as to avoid detection by spreading out the

transfers or by trying to mask the origin of the funds through a centralized point (Ethena: Mint redeem v2). The repeated usage of the same receiving address suggests a consolidation of funds, which could then be laundered further.

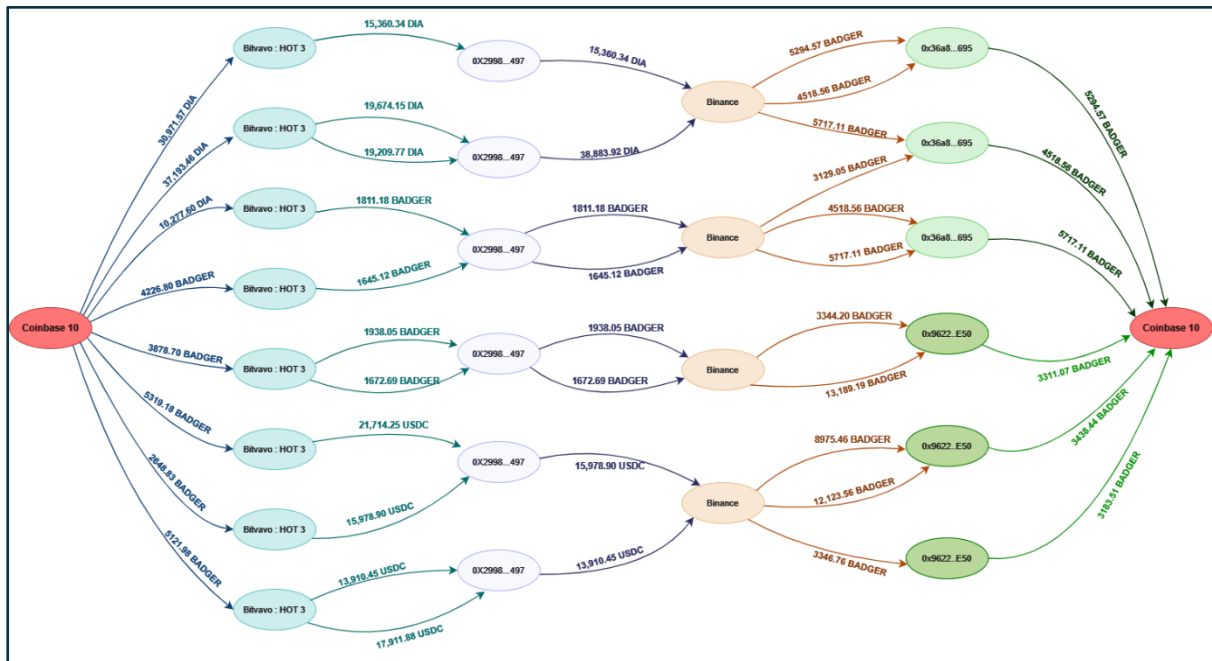


Figure 5.8: *Suspicious Token Transfer Pattern 2*

The **Figure 5.8** shows a more complex series of transactions starting with "Coinbase 10." This entity sends various assets, including DIA (a decentralized oracle asset), BADGER (a DeFi token), and USDC (a stablecoin), to multiple intermediate wallets (such as Bitvavo: HOT 3, and an address starting with "0xC2998...497"). These intermediate wallets then forward the funds to several Binance-associated wallets.

The Binance wallets further redistribute the assets back to other wallets, including the original "Coinbase 10" wallet. This circular flow of funds, with assets being shuffled between multiple intermediaries and then returned to the original source, is highly indicative of a layering technique in money laundering. The purpose here could be to obfuscate the trail of funds, making it difficult for investigators to trace the money back to its original source.

5.7 Discussion

This chapter presents the outcomes from executed four scripts that are designed for interacting with the Ethereum blockchain, focusing on ERC-20 token transfers. Initially, the transaction data extraction script successfully retrieved detailed logs using the Ethereum JSON-RPC API, capturing essential transaction details that are transaction hashes, sender and receiver addresses, token amounts, and contract addresses. Validation of this data against QuickNode confirmed its accuracy, ensuring consistency across nodes.

Subsequently, the validated data was exported to Excel, organizing it into a structured format for easier analysis. This included key columns like transaction hash, sender and receiver addresses, token amounts, and contract addresses, facilitating detailed tracking and filtering of transactions.

The visualization script employed Plotly and Dash Cytoscape to create dynamic and interactive graphs of token transfer networks. These visualizations highlighted key participants.

Finally, the identification of suspicious token transfers focused on manually detecting anomalies that are unusually large transfers, and frequent transactions between specific addresses. These criteria helped to pinpoint potential money laundering or fraudulent activities, providing valuable insights for further investigation. The results underscore the effectiveness of combining data extraction, validation, visualization, and analysis techniques to monitor and investigate Ethereum blockchain transactions comprehensively.

Chapter 6

6. Evaluation

6.1 Introduction

This chapter provides a comprehensive evaluation of the methodologies, tools, and scripts used in the project to interact with the Ethereum blockchain, focusing on their accuracy, efficiency, cost-effectiveness, and overall effectiveness. The evaluation is supported by evidence and metrics that critically analyse the project's success, ensuring that the methods employed meet the project's objectives.

6.2 Data Validation and Accuracy

The accuracy of the data retrieved from the Ethereum blockchain was a critical component of this project. The validation process compared data obtained from the local E-Node with that from QuickNode. Metrics used for validation included transaction hashes, token amounts, and Ethereum addresses.

The **Table 1.** highlights four validated transactions, showcasing a 100% match across all metrics. This includes:

- **Transaction Hashes:** Each transaction's unique identifier was compared between E-Node and QuickNode. All hashes were identical, indicating that both nodes recognized the same transactions.
- **Sender and Receiver Addresses:** For each transaction, the sender and receiver addresses were validated to confirm the correct flow of assets. Both nodes showed identical addresses, further supporting data integrity.
- **Token Amounts:** The amounts of tokens transferred during each transaction, such as BADGER and USDC, were precisely the same across both nodes, ensuring that the financial data was accurately captured.

	Validation Aspect	E-Node Data	QuickNode Data	Validation Results
1	Transaction Hashes	0x4606654aaf02a0c69b57e10d7ddc0cbd98d23e5d681cfe60a5a415cc4ba04986	0x4606654aaf02a0c69b57e10d7ddc0cbd98d23e5d681cfe60a5a415cc4ba04986	✅ 100% Match
	Sender Address	0x36a8dCDbF5F1Fdbf559b79a3fBb3791EEDeDC695	0x36a8dCDbF5F1Fdbf559b79a3fBb3791EEDeDC695	✅ 100% Match
	Receiver Address	0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43	0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43	✅ 100% Match
	Token Amount	5294.57 BADGER (Badger)	5294.57 BADGER (Badger)	✅ 100% Match
2	Transaction Hashes	0xf0ca92d76d405286b4aa231e3685b1057745b2b59d05b86335511d9aa4a9e34f	0xf0ca92d76d405286b4aa231e3685b1057745b2b59d05b86335511d9aa4a9e34f	✅ 100% Match
	Sender Address	0x21a31Ee1afC51d94C2eFcCAa2092aD1028285549	0x21a31Ee1afC51d94C2eFcCAa2092aD1028285549	✅ 100% Match
	Receiver Address	0x36a8dCDbF5F1Fdbf559b79a3fBb3791EEDeDC695	0x36a8dCDbF5F1Fdbf559b79a3fBb3791EEDeDC695	✅ 100% Match
	Token Amount	5294.5 BADGER (Badger)	5294.5 BADGER (Badger)	✅ 100% Match

3	Transaction Hashes	0xae6a8a9dda931fe8eeac8f194723d57cf970c546c04a0c16f216bfc8f1c71d20	0xae6a8a9dda931fe8eeac8f194723d57cf970c546c04a0c16f216bfc8f1c71d20	✓ 100% Match
	Sender Address	0x36a8dCDbF5F1Fdbf559b79a3fBb3791EEDeDC695	0x36a8dCDbF5F1Fdbf559b79a3fBb3791EEDeDC695	✓ 100% Match
	Receiver Address	0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43	0xA9D1e08C7793af67e9d92fe308d5697FB81d3E43	✓ 100% Match
	Token Amount	5303.61 BADGER (Badger)	5303.61 BADGER (Badger)	✓ 100% Match
4	Transaction Hashes	0x9820fce81e5ad1a7d963be407bceadeb5d000a07a8b08d468030748c413e6774	0x9820fce81e5ad1a7d963be407bceadeb5d000a07a8b08d468030748c413e6774	✓ 100% Match
	Sender Address	0x1980475c3a279935Ca611c9A41f17974ebC2e050	0x1980475c3a279935Ca611c9A41f17974ebC2e050	✓ 100% Match
	Receiver Address	0x28104d4F703EE5B5011cefe106f54eFd56F33f95	0x28104d4F703EE5B5011cefe106f54eFd56F33f95	✓ 100% Match
	Token Amount	71947.988805 USDC	71947.988805 USDC	✓ 100% Match

Table 1: Validated Transaction Data

The validation confirms that the data extraction methods are reliable and accurate. The use of two independent nodes (E-Node and QuickNode) as verification sources significantly reduces the likelihood of data errors. This step is crucial for ensuring the integrity of the subsequent analysis, such as the identification of suspicious transactions.

The validation process confirms that the transaction details are consistent and accurate across two independent nodes, significantly reducing the likelihood of data errors. This step is crucial for ensuring the integrity of subsequent analyses, such as the identification of suspicious transactions. By confirming the reliability of the data through the validation process, the project can confidently proceed with further analysis.

6.3 Performance Analysis of Data Extraction Scripts

The performance of the data extraction scripts was evaluated based on the time taken to retrieve transaction logs from the Ethereum blockchain under varying workloads.

The performance analysis revealed a linear increase in processing time as the volume of transaction logs increased. This indicates that while the scripts scale predictably, there may be potential obstruction that could affect performance with extremely large datasets.

The performance was notably influenced by the laptop's specifications, particularly the CPU, RAM, and storage. The tests, conducted on a laptop with an Intel Core i7-9750H CPU, 16 GB RAM, and 512 GB SSD, [63] showed that the scripts performed efficiently under these conditions.

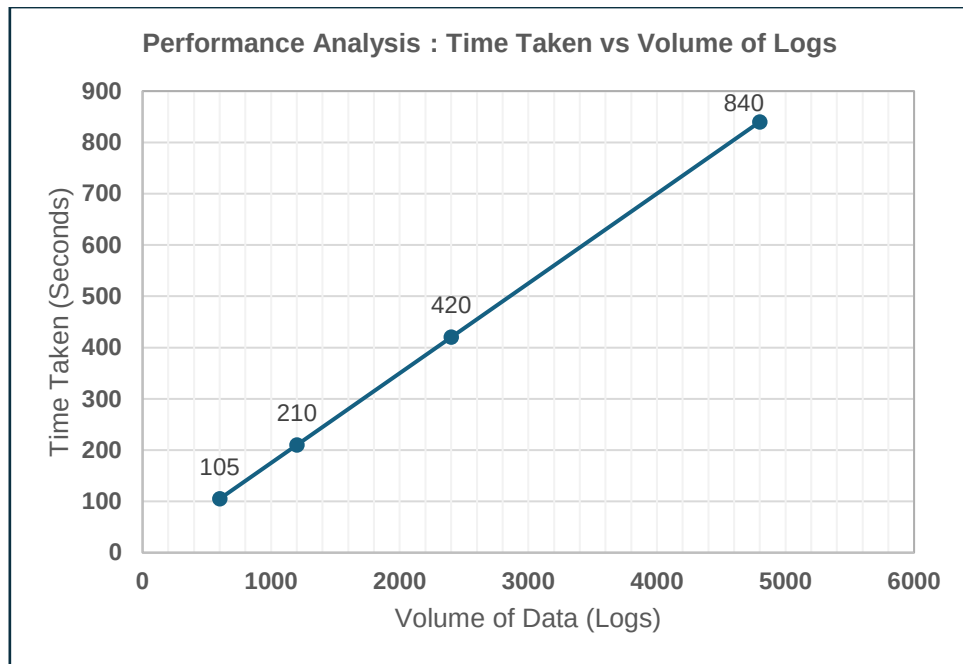


Figure 6.1: Performance Analysis

Figure 6.1 illustrates the correlation between data volume and processing time, confirming the linear scalability of the scripts.

The linear performance suggests that while the scripts are effective for medium-scale data extraction, they may require optimization for handling larger datasets. Future work could explore parallel processing or more efficient data handling techniques to improve performance for extensive data operations.

6.4 Cost-Effectiveness: E- Node vs. QuickNode

The cost analysis in **Figure 6.2**, compared the long-term financial implications of running a personal Ethereum node against using QuickNode.

- **E-Node (Personalized Node):** The initial investment includes hardware costs, which amount to approximately £2000, along with ongoing electricity and maintenance expenses, bringing the total to about £2,535 for the first year. In subsequent years, the costs decrease significantly as the major expenses are already covered, making this option more cost-effective over time. By the third year, the cumulative costs stabilize, showing that running a personal node becomes increasingly economical.
- **QuickNode:** This service offers convenience and scalability at an annual cost of £3,782 [48]. While attractive for users needing quick deployment and minimal maintenance, the cumulative costs rise significantly over time. By the third year, the cumulative cost reaches approximately £11,347, highlighting the higher long-term expenses compared to a personal node.

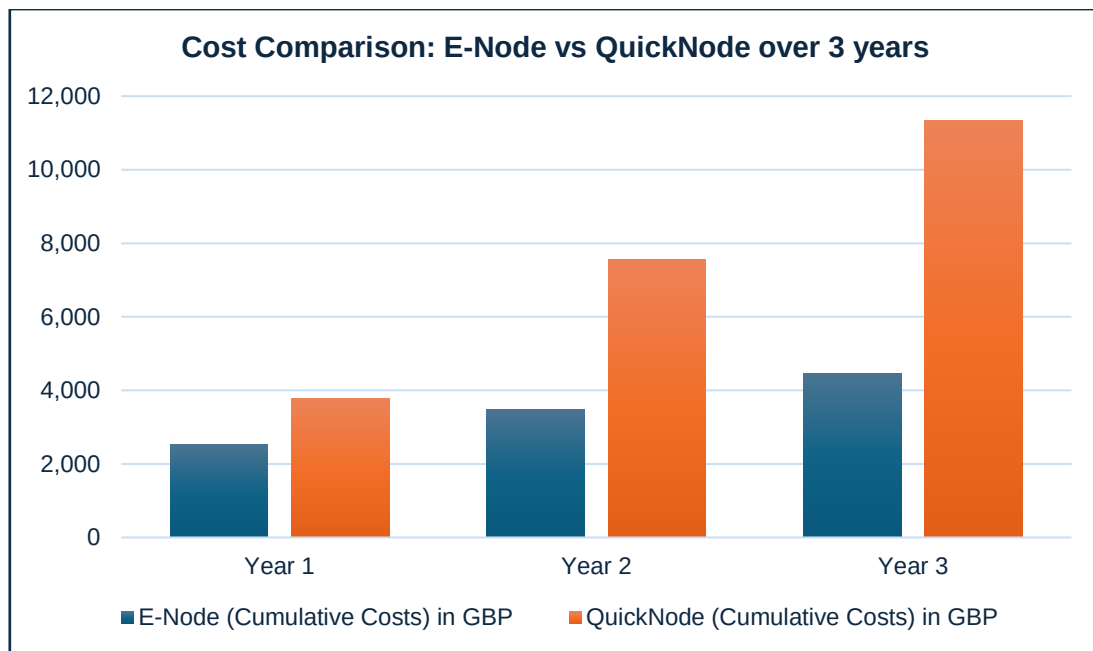


Figure 6.2: *Cost Comparison: E-Node vs QuickNode over 3 years*

The decision between using a personal node or QuickNode should be based on the specific needs of the user. For short-term projects or those requiring rapid deployment, QuickNode offers significant advantages. However, for long-term projects with large data requirements, the cost savings and control offered by a personal node are substantial. The analysis emphasizes the importance of aligning the choice of infrastructure with the project's scale and duration.

6.5 Effectiveness of Visualization Tools

The effectiveness of Plotly and Dash Cytoscape in visualizing ERC-20 token transfers was evaluated based on their ability to represent complex networks clearly and interactively. These tools excelled in mapping token transfer networks, effectively highlighting central nodes and transaction flows. They translated complex data into understandable formats, making it easier to interpret and analyze.

The **Figure 5.5 and 5.6** in the results section demonstrate this effectiveness, capturing key elements of the visualizations, such as the flow of transactions between addresses and the identification of central nodes. The interactive nature of these tools is particularly valuable for uncovering patterns and anomalies that might not be apparent through raw data analysis alone. This graphical interaction enhances the overall analysis, providing a clear and accessible way to identify potential suspicious transactions and gain deeper insights into the network dynamics.

6.6 Identification of Suspicious Token Transfers

The results demonstrated in **Section 5.6** shows that the manual tracing of suspicious activity through Ethereum's JSON-RPC API was successful in uncovering transactions indicative of illicit activities. These flagged transactions provided actionable insights, which can serve as starting points for deeper investigation into potential money laundering or fraudulent activities within DeFi ecosystems.

The suspicious transactions revealed unusual patterns in token movements, such as large sums transferred in short periods, and circular flow of funds which often correlate with efforts to hide illicit activity, such as layering and smurfing techniques used in money laundering. The manual methodology employed effectively captured these patterns by focusing on tracing specific ERC-20 token transfers that exhibited characteristics associated with money laundering.

Despite the success of manual detection, the evaluation highlighted areas where the process could be significantly improved through automation. The manual process, while accurate, is time-consuming and may become increasingly impractical as the volume of blockchain transactions continues to grow. Automation, particularly through machine learning and anomaly detection algorithms, could provide a more efficient and scalable approach to identifying suspicious token transfers.

6.7 Discussion

The assessments in the sections 6.2 and 6.4 reveals the robustness and practicality of the project's approach respectively. The validation process, which compared data from the local E-Node with QuickNode, demonstrated a 100% match across key metrics such as transaction hashes, sender and receiver addresses, and token amounts, confirming the accuracy of the extraction methods. This reliability is critical for ensuring the integrity of further analysis, particularly in identifying suspicious activities.

From a cost perspective, running a personal Ethereum node proved more economical in the long term. While QuickNode offers immediate deployment and scalability, its cumulative costs make it less viable for extended projects. E-Node, despite higher initial expenses, offers significant cost savings over time, making it the more sustainable option for ongoing transaction monitoring.

The evaluation of Section 6.3 and Section 6.5 further highlights the system's efficiency and analytical capabilities. The performance analysis demonstrated linear scalability of the data extraction scripts, with processing time increasing proportionally as transaction volumes grew. This indicates that while the system functions effectively for medium-scale operations, further optimization could enhance its performance with larger datasets, ensuring it remains responsive in high-transaction environments.

The visualization tools Plotly and Dash Cytoscape, proved highly effective in mapping ERC-20 token transfers, offering clear and interactive visual representations of complex transaction networks. These tools enabled the identification of central nodes and transaction patterns critical to detecting illicit financial activities, significantly improving the system's ability to analyze and interpret large datasets.

Manual analysis using the Ethereum JSON-RPC API successfully identified suspicious token movements, such as large sums transferred over short periods and circular fund flows indicative of money laundering techniques like layering and smurfing. However, the manual process, while successful, is time consuming and not scalable for large volumes of blockchain transactions.

To address these limitations, automating the detection process through advanced anomaly detection algorithms presents an effective alternative. Automation, driven by machine learning techniques, offers several advantages over manual methods. Incorporating machine learning models, particularly supervised learning trained on historical data, would allow the system to detect suspicious patterns, such as rapid token transfers between new wallets, more

efficiently. This automation would not only reduce the need for manual intervention but also significantly enhance the system's ability to handle large transaction volumes in real-time.

Moreover, automated systems can address complex challenges such as detecting transactions involving mixers like TornadoCash. Mixers obscure transaction origins, making it difficult to trace fund flows through traditional methods. Machine learning models can be designed to recognize patterns typical of mixer transactions, such as fragmented transfers or frequent small transactions. By focusing on the flow of funds to cryptocurrency exchanges, these models can enhance the ability to detect and investigate suspicious activities effectively.

In addition to supervised learning, unsupervised learning techniques, such as clustering algorithms, can be employed to identify anomalies by grouping transactions based on attributes like size, frequency, and timing. When integrated with rule-based systems that target specific money laundering behaviors, these techniques can improve the accuracy and efficiency of detecting suspicious transactions. Automated systems can thus offer a more comprehensive approach to anomaly detection, combining various algorithms to capture a wide range of illicit activities.

Chapter 7

7. Conclusion

This dissertation demonstrates that the Ethereum JSON-RPC API can serve as a powerful, cost-effective tool for monitoring and analysing blockchain transactions, particularly in identifying and mitigating illicit activity of money laundering within DeFi ecosystems. By employing the standard API, this research offers a viable alternative to costly third-party service, QuickNode, providing a more accessible approach to forensic analysis and transaction tracing.

The findings underscore the potential of leveraging publicly available blockchain data and APIs to enhance transparency and regulatory compliance in decentralized environments. Through effective use of the JSON-RPC API, it is possible to retrieve and decode detailed transaction data, visualize complex token transfer networks, and identify suspicious patterns indicative of illicit financial flows. This approach aligns with the need for scalable and practical solutions that can operate within the constraints of decentralized financial systems while still addressing security and regulatory challenges.

In conclusion, this dissertation provides extracted data of traced and monitored ERC-20 token and analysed blockchain transactions to detect illicit activity of money laundering, within DeFi ecosystem by utilising Ethereum's JSON-RPC API methods. Future research could further develop these methodologies, including integrating machine learning techniques and advanced analytics, to automate the detection of suspicious activities and improve the scalability and accuracy of transaction monitoring. This work contributes to the broader understanding of how blockchain technology can be harnessed to support a secure, transparent, and compliant financial ecosystem, balancing the benefits of innovation with the imperatives of regulation and oversight.

Chapter 8

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Appendix

9.1 Git Repository

Address of git repository: <https://git.cs.bham.ac.uk/projects-2023-24/jxg447>

The git repository contains four Python programs for data extraction, data validation, data exportation to Excel and graphical visualization.